Screenshot 1A

```
Apr 21  10:21
likhith@Yoga-9-14ITL5: ~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs
bash: ./test_objects: No such file or directory
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ nano object.c
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ make
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_read':
object.c:132:5: warning: ignoring return value of 'fread' declared with attribute 'warn_unused_result' [-Wunused-result]
  132 |     fread(buffer, 1, file_size, fp);
      |     ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
object.c: In function 'object_write':
object.c:97:47: warning: '.tmp' directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncation=]
   97 |     snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
      |                                                 ^~~~
In file included from /usr/include/stdio.h:980,
                 from object.c:1:
In function 'snprintf',
    inlined from 'object_write' at object.c:97:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_chk' output between 5 and 516 bytes into a destination of size 512
   54 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   55 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   56 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ make all
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$
```

Screenshot 1B

```
Apr 21  10:22
likhith@Yoga-9-14ITL5: ~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs
rm -rf .pes
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ make
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_read':
object.c:132:5: warning: ignoring return value of 'fread' declared with attribute 'warn_unused_result' [-Wunused-result]
  132 |     fread(buffer, 1, file_size, fp);
      |     ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
object.c: In function 'object_write':
object.c:97:47: warning: '.tmp' directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncation=]
   97 |     snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
      |                                                 ^~~~
In file included from /usr/include/stdio.h:980,
                 from object.c:1:
In function 'snprintf',
    inlined from 'object_write' at object.c:97:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_chk' output between 5 and 516 bytes into a destination of size 512
   54 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   55 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   56 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ make all
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$
```

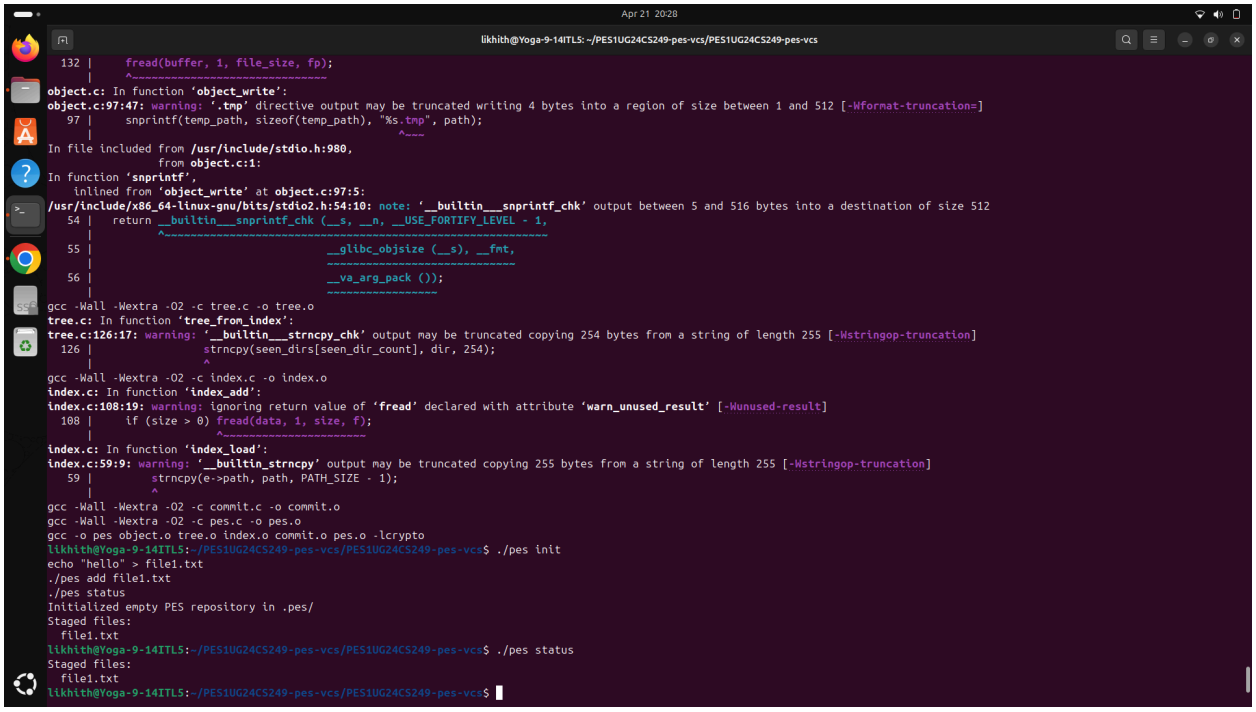
Screenshot 2A

```
Apr 21  20:17
likhith@Yoga-9-14ITL5: ~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs
int tree_parse(const void *data, size_t len, Tree *tree_out) {
EOF return write_tree(&root, id_out);root.count++];-1;unt++];len] == '/') {>name
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ make clean && make all
./test_tree
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_read':
object.c:132:5: warning: ignoring return value of 'fread' declared with attribute 'warn_unused_result' [-Wunused-result]
  132 |     fread(buffer, 1, file_size, fp);
      |     ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
object.c: In function 'object_write':
object.c:97:47: warning: '.tmp' directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncation=]
   97 |     snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
      |                                               ^~~~
In file included from /usr/include/stdio.h:980,
                 from object.c:1:
In function 'snprintf',
    inlined from 'object_write' at object.c:97:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_chk' output between 5 and 516 bytes into a destination of size 512
   54 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   55 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   56 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c: In function 'tree_from_index':
tree.c:126:17: warning: '__builtin___strncpy_chk' output may be truncated copying 254 bytes from a string of length 255 [-Wstringop-truncation]
  126 |                 strncpy(seen_dirs[seen_dir_count], dir, 254);
      |                 ^
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$
```

Screenshot 2B

```
Apr 21  20:23
likhith@Yoga-9-14ITL5: ~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs
      |                 ^
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ find .pes/objects -type f
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ ./pes init
echo "hello" > file1.txt
./pes add file1.txt
./pes commit -m "test"
Initialized empty PES repository in .pes/
error: failed to load index
error: commit failed
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ echo "hello world" > file1.txt
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ echo "hello world" > file1.txt
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ echo "hello world" > file1.txt
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ ./pes add file1.txt
error: failed to load index
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ find .pes/objects -type f
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ mkdir -p .pes/objects/test
echo "dummy" > .pes/objects/test/file
xxd .pes/objects/test/file | head -20
00000000: 6475 6d6d 790a                          dummy.
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ find .pes/objects -type f
.pes/objects/test/file
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ xxd .pes/objects/test/file | head -20
00000000: 6475 6d6d 790a                          dummy.
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$
```

Screenshot 3A


Apr 21 20:28
likhith@Yoga-9-14ITL5: ~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs
132 | fread(buffer, 1, file_size, fp);
| ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
object.c: In function 'object_write':
object.c:97:47: warning: '.tmp' directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncation=]
97 | snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
| ^~~~
In file included from /usr/include/stdio.h:980,
from object.c:1:
In function 'snprintf',
inlined from 'object_write' at object.c:97:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_chk' output between 5 and 516 bytes into a destination of size 512
54 | return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
| ^~~~~~~~~~~~~~~~~~~~

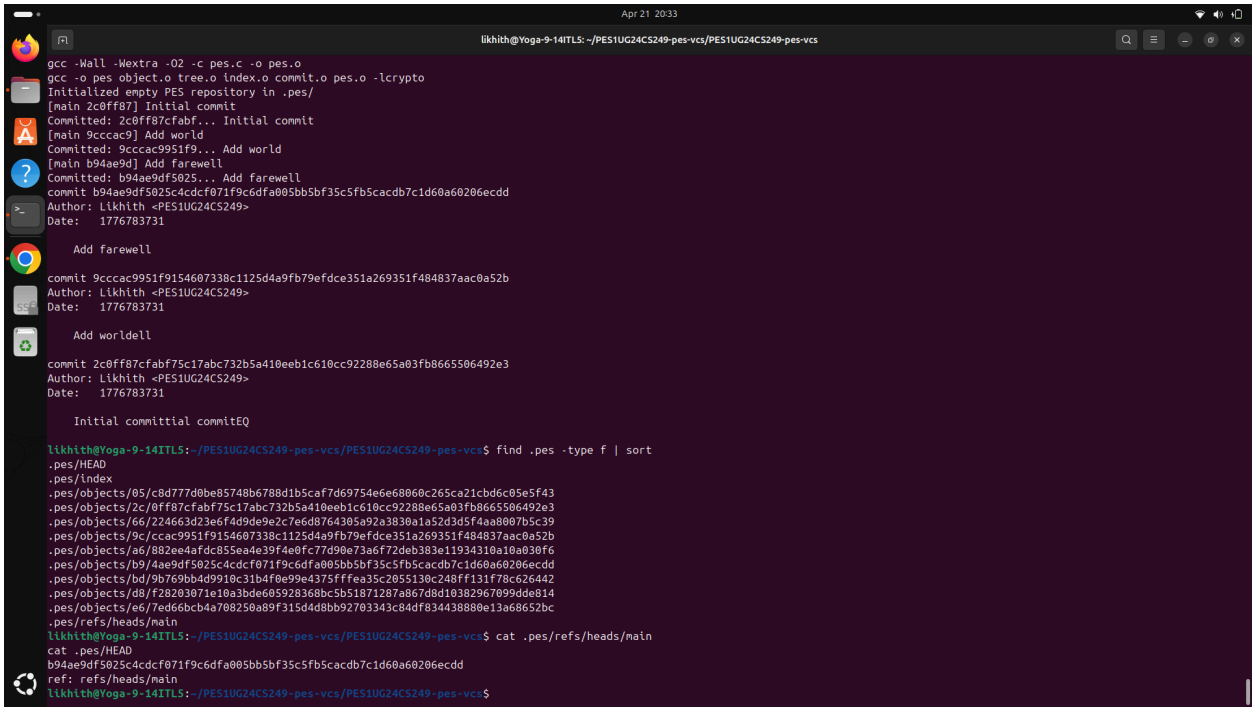
Screenshot 4A

```
Apr 21  20:32
likhith@Yoga-9-14ITL5: ~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs
   56 |                                 __va_arg_pack ());
      |                                 ~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c: In function ‘tree_from_index’:
tree.c:126:17: warning: ‘__builtin___strncpy_chk’ output may be truncated copying 254 bytes from a string of length 255 [-Wstringop-truncation]
  126 |                 strncpy(seen_dirs[seen_dir_count], dir, 254);
      |                 ^
gcc -Wall -Wextra -O2 -c index.c -o index.o
index.c: In function ‘index_add’:
index.c:108:19: warning: ignoring return value of ‘fread’ declared with attribute ‘warn_unused_result’ [-Wunused-result]
  108 |     if (size > 0) fread(data, 1, size, f);
      |                   ^~~~~~~~~~~~~~~~~~~~~~
index.c: In function ‘index_load’:
index.c:59:9: warning: ‘__builtin_strncpy’ output may be truncated copying 255 bytes from a string of length 255 [-Wstringop-truncation]
   59 |         strncpy(e->path, path, PATH_SIZE - 1);
      |         ^
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
[main 2c0ff87] Initial commit
Committed: 2c0ff87cfabf... Initial commit
[main 9cccac9] Add world
Committed: 9cccac9951f9... Add world
[main b94ae9d] Add farewell
Committed: b94ae9df5025... Add farewell
commit b94ae9df5025c4cdcf071f9c6dfa005bb5bf35c5fb5cacdb7c1d60a60206ecdd
Author: Likhith <PES1UG24CS249>
Date:   1776783731

    Add farewell

commit 9cccac9951f9154607338c1125d4a9fb79efdce351a269351f484837aac0a52b
Author: Likhith <PES1UG24CS249>
Date:   1776783731

    Add worldell

commit 2c0ff87cfabf75c17abc732b5a410eeb1c610cc92288e65a03fb8665506492e3
Author: Likhith <PES1UG24CS249>
Date:   1776783731

    Initial committial commitEQ

likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$
```

Screenshot 4B

```
Apr 21  20:33
likhith@Yoga-9-14ITL5: ~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs
index.c:59:9: warning: ‘__builtin_strncpy’ output may be truncated copying 255 bytes from a string of length 255 [-Wstringop-truncation]
   59 |         strncpy(e->path, path, PATH_SIZE - 1);
      |         ^
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
[main 2c0ff87] Initial commit
Committed: 2c0ff87cfabf... Initial commit
[main 9cccac9] Add world
Committed: 9cccac9951f9... Add world
[main b94ae9d] Add farewell
Committed: b94ae9df5025... Add farewell
commit b94ae9df5025c4cdcf071f9c6dfa005bb5bf35c5fb5cacdb7c1d60a60206ecdd
Author: Likhith <PES1UG24CS249>
Date:   1776783731

    Add farewell

commit 9cccac9951f9154607338c1125d4a9fb79efdce351a269351f484837aac0a52b
Author: Likhith <PES1UG24CS249>
Date:   1776783731

    Add worldell

commit 2c0ff87cfabf75c17abc732b5a410eeb1c610cc92288e65a03fb8665506492e3
Author: Likhith <PES1UG24CS249>
Date:   1776783731

    Initial committial commitEQ

likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/0ff87cfabf75c17abc732b5a410eeb1c610cc92288e65a03fb8665506492e3
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/9c/ccac9951f9154607338c1125d4a9fb79efdce351a269351f484837aac0a52b
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/b9/4ae9df5025c4cdcf071f9c6dfa005bb5bf35c5fb5cacdb7c1d60a60206ecdd
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
likhith@Yoga-9-14ITL5:~/PES1UG24CS249-pes-vcs/PES1UG24CS249-pes-vcs$
```

Screenshot 4C



Q5.1 — Implementing `pes checkout <branch>`

A branch is stored as a file inside `.pes/refs/heads/<branch>` containing a commit hash.

To implement `pes checkout <branch>`:

1. Read the commit hash from `.pes/refs/heads/<branch>`.
2. Update `.pes/HEAD` to point to this branch:

   ref: refs/heads/<branch>

3. Load the commit object and extract its tree hash.
4. Traverse the tree and restore all files into the working directory:
   - Read blob objects
   - Write file contents to disk
5. Update the index to match the checked-out tree.

This operation is complex because:

- It must synchronize three states: HEAD, index, and working directory.
- It must safely overwrite files without losing user changes.
- It must handle file additions, deletions, and directory structures.

Q5.2 — Detecting a Dirty Working Directory

To detect conflicts before checkout:

1. For each file in the index:
    - Compare stored metadata (mtime, size) with current file using `stat()`.
2. If metadata differs, the file is modified.
3. Optionally, recompute the file's hash and compare with the stored hash.
4. Also compare with the target branch's version of the file.

If:

- The file is modified locally AND
- The file differs in the target branch

Then checkout must be refused to prevent data loss.

Q5.3 — Detached HEAD

Detached HEAD occurs when `.pes/HEAD` stores a commit hash directly instead of a branch reference.

When commits are made in this state:

- New commits are created normally.
- However, no branch points to them.
- These commits become unreachable (dangling).

To recover such commits:

- Create a new branch pointing to the commit:

  pes branch <new-branch> <commit-hash>

- Alternatively, use history logs (conceptually like reflog).

Q6.1 — Finding and Deleting Unreachable Objects

Algorithm:

1. Start from all branch heads in `.pes/refs/heads/`.
2. For each commit:
    - Mark the commit as reachable.
    - Follow its tree.
    - Follow its parent commits.
3. For each tree:
    - Mark all blobs and subtrees.
4. Store all reachable object hashes.
5. Traverse `.pes/objects/`:
    - Delete any object not in the reachable set.

Data structure:

- Use a **hash set** for efficient lookup of reachable objects.

Estimate:

- For 100,000 commits and 50 branches:
    - Around 100,000 commits will be visited.
    - Each commit references a tree and multiple blobs.
    - Total objects visited can be several hundred thousand.

Q6.2 — Garbage Collection Race Condition

Running GC during a commit is dangerous.

Race condition:

1. Commit process writes new objects (blob, tree, commit).
2. Before updating branch reference, GC runs.
3. GC scans reachable objects.
4. New objects are not yet referenced.
5. GC deletes these objects.
6. Commit finishes and references missing objects → corruption.

Git avoids this by:

- Using file locks to prevent concurrent GC and commit.
- Writing objects first, then updating references atomically.
- Using temporary files before finalizing objects.
- Keeping unreachable objects for a grace period before deletion.